



**THE UNIVERSITY
OF QUEENSLAND**
A U S T R A L I A

Fear of temperature

Computational analysis of policy, media
and public climate emotions

Project proposal

Dai Pan

Supervisor: Mashhuda Glencross

REIT7842

16 September 2026

Use of AI Statement

Topic exploration: Yes; AI supported discussion of the topic and candidate approaches.

Research questions: Yes; AI assisted formulation and critique of Section 3.1; I selected the research focus.

Literature review: Yes; AI assisted searches and selected claim/reference checks for Section 2. Reading depth is recorded separately.

Formatting and language: Yes; AI assisted English drafting, translation, editing and document layout.

Other uses: Methodological discussion, planning, Figures 1-8 and export code. No empirical research-data analysis is reported here.

I acknowledge the use of generative AI tools in completing this assessment. Details of which tools were used and how they were used are provided in the table below, along with appropriate in-text and full references. I take responsibility for critically evaluating and integrating the AI-generated content, and ensuring it adheres to academic integrity standards.

Table A. AI tool use.

Tool and recorded use	Main activities
ChatGPT; 11 and 15 September 2026	Topic and question development, literature assistance, methodological discussion, writing and language editing.
Gemini; date/version unrecorded	Suggested NLP modules and event/lag analysis; suggestions were assessed and revised by the author.
Codex; 15 September 2026	Source and feasibility checks, discussion records, drafting, figures, export code and layout inspection.

Tools: ChatGPT ([OpenAI, n.d.-a](#)), Gemini ([Google, n.d.](#)), Codex ([OpenAI, n.d.-b](#)). Model versions are retained in the working record where available. This declaration concerns proposal preparation; it reports no empirical research-data analysis.

Contents

1. Introduction	1
1.1 Motivation and significance	1
1.2 A methodological transition	1
1.3 Scope and historical framing	2
1.4 Broad aim and intended contribution	2
2. Background and Literature Review	3
2.1 Climate emotions and anticipated futures	3
2.2 Agenda-setting and responsibility frames	3
2.3 Evidence from climate concern and role comparisons	4
2.4 Competing temporal mechanisms and expectations	4
2.5 NLP for emotion, relations and temporal text analysis	5
2.6 Selected research gap	5
3. Project Plan	6
3.1 Research questions	6
3.2 Evidence-driven implementation	6
3.3 Corpus construction and sampling	7
3.4 Operational definitions and construct validity	8
3.5 Representation and three NLP components	10
3.6 Temporal alignment and analysis	11
3.7 Evaluation and validity	13
3.8 Project risks and scope adjustment	14
3.8.1 Risk assessment	14
3.8.2 Review points and scope adjustment	14
3.9 Milestones, resources and time allocation	15
3.10 OHS, ethics and data governance	16
3.11 Outputs, interpretation and research record	16
References	17

Figures and tables

Figures: 1. Research relationships; 2. Competing mechanisms; 3. RQ evidence routes; 4. Source denominators; 5. NLP pipeline; 6. Measurement and timing; 7. Contextual evidence; 8. Calendar and workload.

Tables: A. AI use; 1. Literature synthesis; 2. Source register; 3a-b. Measures and annotation; 4. Event plan; 5. Evaluation; 6. Risk assessment; 7. OHS and ethics.

Reading note. This proposal describes planned research; it does not report completed experiments. Numerical examples and diagram trajectories explain the design. Access-dependent choices will be resolved through the documented pilot. The provisional submission date is 16 September 2026; the course deadline is 17 September 2026 at 15:00.

1. Introduction

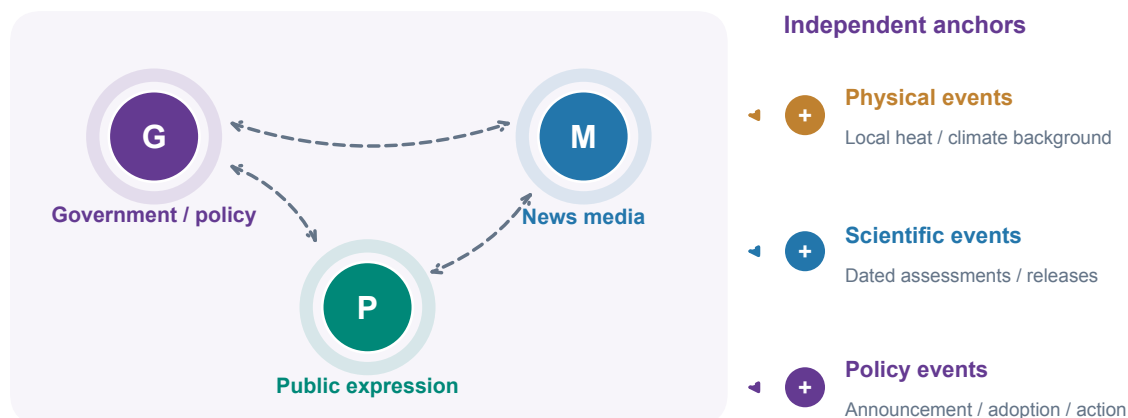
1.1 Motivation and significance

Rising temperature is both a physical process and an object of social anticipation. People may describe immediate danger during a heatwave, worry about the world their children will inherit, or assign responsibility for future disruption. These expressions involve different emotions, time horizons and social relationships. Climate-anxiety research recognises psychological responses to anticipated environmental change as well as direct exposure (Clayton, 2020). Studying how such responses are communicated can illuminate the changing social meaning of warming without assuming that every negative statement expresses fear.

Fear of Temperature asks how warming-related fear and anxiety develop across government and policy discourse, news media and public expression. A central problem is temporal ordering: does public concern follow physical events, institutional communication or media attention, and when might it precede them? Earlier work links climate concern to political cues, media and other conditions (Brulle et al., 2012). Research on affective agenda dynamics also demonstrates the relevance of comparing government, media and public expressions (Zhou et al., 2023). Figure 1 frames these as competing relationships to investigate rather than a predetermined chain of influence.

One social question, three discourse roles

Who changes first, and how is warming-related fear explained?



Dashed arrows: competing relationships, not established effects.

Figure 1. Three discourse roles. Dashed links are candidate relationships, not causal findings.

1.2 A methodological transition

This computational social science study investigates climate communication through a reproducible corpus, an NLP pipeline and temporal analysis. Earlier dictionary and n-gram exploration develops into natural-language collection, database construction and Transformer representations. We compare computational measurements against explicit baselines and contextual evidence, reducing discretionary passage selection while testing construct validity. The purpose remains the author's investigation of social emotions at scale; the principal scholarly outcome is the thesis.

1.3 Scope and historical framing

The core collection will prioritise English-language texts associated with the United States, European settings, Australia and New Zealand. Broad geographical coverage is an objective, while the main comparison concerns source roles and identifiable communities.

Chinese-language material, East Asian settings and AOSIS members are extension candidates rather than dependencies of the core study. Publication location, author location and the place discussed in a text will be distinguished. English language alone will not establish geographic or cultural identity.

The target collection begins in January 1988 and extends to a documented cutoff in 2026. The establishment of the IPCC in 1988 motivates an institutional anchor ([Intergovernmental Panel on Climate Change, n.d.](#)). It does not establish an emotional turning point. Recoverable material from 1938 onwards may form an earlier contextual layer, with Callendar's work providing a scientific-historical reference ([Callendar, 1938](#)). A claim about change across 1988 requires adequate observations on both sides; the main collection alone cannot demonstrate that transition.

Historical span and analytical coverage are therefore different. Each comparison will use a sufficiently dense common window, with incomplete months, archive gaps and changes of source explicitly recorded. Letters to editors and early forums may extend the record of public expression, but their selection processes differ from contemporary platforms. The intended inference concerns the sampled discourse. Claims about population emotion require additional sampling or validation evidence.

Longer-term anticipated consequences are the substantive priority. Direct heat danger will be collected alongside them and labelled after collection. Emotional category, immediate bodily-threat cues, time horizon and threatened group will be separate, potentially overlapping attributes. Fear can concern the future, and collective anxiety can arise during a present emergency. Cultural heritage may inform discussion, but the project does not presuppose heritage status.

1.4 Broad aim and intended contribution

The broad aim is to understand how warming-related social emotions, anticipated futures and responsibility narratives change across policy, news and public discourse, and how their temporal relationships vary with physical and institutional events.

Technical resource. A versioned, traceable corpus and reproducible pipeline with documented coverage, denominators, sampling weights and a data dictionary. Code, configurations and random seeds will support reproduction; only permitted material will be released.

Validated method. A modular workflow connecting relevance detection, emotion, relations and topics to temporal analysis, evaluated against baselines and across sources and periods. Any algorithmic novelty claim depends on demonstrated improvement, not simply combining existing models.

Empirical contribution. A thesis explaining temporal relationships and responsibility narratives through statistical estimates and contextual evidence, with explicit causal limits and potential for publication. Small or uncertain associations remain legitimate findings; success does not require a dominant driver or significant policy effect.

2. Background and Literature Review

2.1 Climate emotions and anticipated futures

Climate anxiety concerns responses to possible as well as experienced environmental harm. Clayton cautions against assuming that concern is necessarily maladaptive (Clayton, 2020). Solastalgia instead describes distress associated with environmental change in a place to which people remain attached (Albrecht et al., 2007). These accounts connect emotion to anticipated futures and lived environments, but neither makes all negative climate language equivalent to anxiety.

Pihkala's taxonomy distinguishes fear, worry, grief, anger, guilt and hope (Pihkala, 2022). For this project, that diversity argues against a single scale running from bodily panic to existential anxiety. Emotional category, target, time horizon and affected group need separate labels. Textual worry does not establish a persistent psychological condition; keyword cues require contextual validation.

O'Neill and Nicholson-Cole examine fear-inducing climate imagery and engagement (O'Neill & Nicholson-Cole, 2009). Their distinction between attracting attention and supporting engagement matters here: a rise in threatening representations need not indicate stronger policy support. The proposed measurements therefore separate issue attention, emotional expression and stance. This gives a more specific account of discourse than a negative-sentiment score, while retaining the limitation that expressions are not direct observations of population mental health.

2.2 Agenda-setting and responsibility frames

Agenda-setting addresses relationships between the prominence of issues in media and public agendas (McCombs & Shaw, 1972). A corpus share, however, measures the prominence of an issue in sampled output; without audience information it does not measure exposure or establish whose agenda changed.

Framing concerns how communication selects a problem, causal interpretation, evaluation and response (Entman, 1993). Increased climate coverage can frame warming as economic disruption, physical danger or intergenerational injustice. These are different interpretations even if their overall issue attention is similar. Cause, blame and response duty also differ: an actor may be asked to respond without being blamed for creating the problem. Relations and contextual evidence are therefore needed alongside frequency and emotion measures.

Similar policy and public language may reflect adoption, quotation or opposition; convergence alone does not establish agreement.

Table 1. Literature and design.

Evidence strand	What it contributes	Requirement for this study
Climate emotions	Multiple responses to threat	Separate emotion, target and horizon
Agenda and framing	Prominence versus interpretation	Measure attention and narrative content
Climate concern	Elite/media and physical explanations	Compare roles and exposure scales
Contextual NLP	Retrieval and structured text labels	Validate by source and period
Temporal methods	Ordering and event-related changes	State estimands and causal limits

2.3 Evidence from climate concern and role comparisons

Brulle and colleagues construct quarterly U.S. climate-concern measures from 74 surveys during 2002–2010, comparing weather, science information, media and political factors (Brulle et al., 2012). Their findings motivate institutional explanations, but aggregate concern is not specifically future-oriented fear. National quarterly observations may also conceal responses to local, short-lived heat exposure. The proposed text measures add narrative detail and shorter eligible windows; they introduce selection and classification errors that survey-based concern does not share in the same form. They should complement this evidence rather than be assumed more accurate.

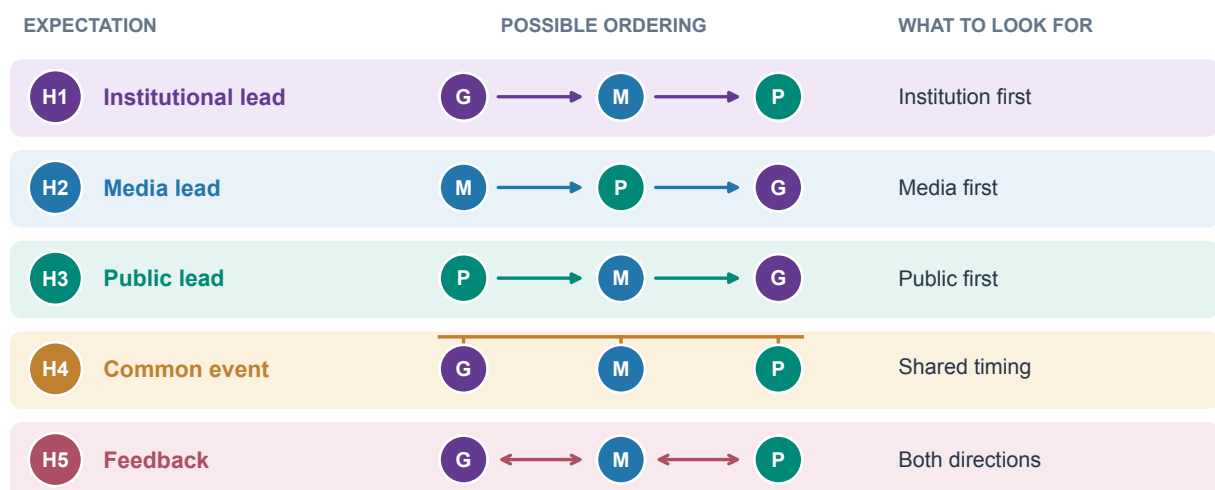
Zhou and colleagues compare public, government and media emotional communication on Weibo during COVID-19 (Zhou et al., 2023). This is a close precedent for role-based temporal comparison, but one platform and a pandemic period differ from a historical climate corpus spanning institutions and media systems. The present design retains the role comparison while testing source continuity, emotional targets and independent physical observations. Neither precedent establishes that a fixed direction of influence applies across all periods.

2.4 Competing temporal mechanisms and expectations

Institutional signals can supply authoritative risk interpretations (Brulle et al., 2012); media can select and frame issues (Entman, 1993; McCombs & Shaw, 1972); public expression may supply news material and pressure for response. Figure 2 makes the competing temporal expectations visible. H3 is a project hypothesis informed by role-comparison research (Zhou et al., 2023), not an established climate result.

Five competing temporal patterns

Read each row as a candidate explanation; routes may differ across windows.



G: government/policy M: media P: public · Predictive ordering alone is not causation.

Figure 2. Competing temporal mechanisms. Arrows indicate candidate orderings; colours identify roles.

The same ordering can arise from anticipation, shared releases, reposting or changing participants. A coarse monthly bin can hide a faster ordering. Local heat exposure also differs from global climate background. These expectations may coexist across windows; temporal prediction alone cannot identify social feedback or individual emotional contagion.

2.5 NLP for emotion, relations and temporal text analysis

Sentence-BERT supports efficient semantic comparison of sentences (Reimers & Gurevych, 2019). This can locate relevant expressions beyond exact dictionary phrases, but similarity does not by itself measure emotion, intensity or agreement. A shared encoder will maintain a common representation; validated relevance decisions will determine which texts enter the climate subset. A lexical/TF-IDF baseline remains useful for assessing what contextual representations recover and which relevant expressions they miss.

Aspect-based sentiment analysis links evaluations to their objects (Pontiki et al., 2014). This matters when a negative statement concerns the cost of a policy rather than warming. GoEmotions distinguishes fine-grained emotions in English Reddit comments (Demszky et al., 2020); it includes fear and nervousness, but no dedicated climate-anxiety category. Neither general sentiment nor an unvalidated model label can supply the project's future-worry measure. Cross-period and cross-source validation will assess this transfer before monthly means are interpreted.

PropBank describes predicate-argument roles (Palmer et al., 2005), while emotion-cause pair extraction links emotional expressions with reported causes (Xia & Ding, 2019). These inform complementary parts of the relation layer. A grammatical subject is not automatically culpable: in "government blamed industry", the speaker and blamed actor differ. The project must also distinguish described causes from attributed blame and response duties, retaining the predicate and evidence span rather than reducing every relation to an unqualified triple.

BERTopic combines embeddings, clustering and class-based term representation (Grootendorst, 2022). It provides a practical topic-analysis candidate, although the cited account is a preprint and suitability for this corpus remains to be tested. A change in topic prevalence can arise from source composition or vocabulary rather than changing meaning. A shared encoder, fixed-source checks, stable clusters and contextual passages are needed before describing semantic or metaphorical evolution.

The three modules jointly connect response and target (emotion), explanation (relations), and recurring narratives (topics). Their combination serves the substantive questions. Agreement is not independent confirmation when modules inherit the same retrieval or representation error.

2.6 Selected research gap

The research gap is a missing connection between role-specific temporal change, the emotion expressed and its attributed cause or responsibility. Attention studies do not by themselves establish whether fear increased, while emotion labels alone do not explain how a threat was framed. This project addresses the gap by joining validated attention, future-oriented emotion and responsibility measures to traceable policy, media and public passages, then comparing their timing within verified common coverage windows. The contribution is an integrated, auditable account of this relationship, not a claim that no previous study combined NLP and climate discourse.

The temporal methods answer bounded questions. Granger analysis concerns added prediction (Shojaie & Fox, 2022); ITS separates level and slope changes (Lopez Bernal et al., 2017), while a temporal cutoff alone does not identify a causal effect (Hausman & Rapson, 2018). Cross-lagged interpretations also require distinguishing stable unit differences from within-unit change (Hamaker et al., 2015). These limitations support the existing small-model pathway, rather than additional methods. The scholarly contribution depends on interpretable evidence and validated measurement, not a predetermined direction of influence.

3. Project Plan

3.1 Research questions

RQ1 - Who changes first? Which roles lead, lag or change together in warming-related attention and emotion? CCF and conditional VAR assess timing and held-out predictive gain over an own-history baseline, accounting for trends, seasonality and serial dependence.

RQ2 - What changes around events? How do heat, scientific and policy events relate to attention, emotion and framing? Segmented regression/ITS estimates level and slope changes, with window sensitivity and placebo-date checks.

RQ3 - How is fear explained? How are causes, blame, duties and threatened futures adopted, contested or reinterpreted across roles? Evidence-linked relations and topics support interpretation; span precision/recall and source/period holdouts test extraction quality.

Future-oriented warming fear or worry in public texts remains the main substantive outcome. The computational tasks make each question testable.

Research questions → computational tasks → evaluation

Three questions share one corpus, representation and validation protocol.

QUESTION	COMPUTATIONAL TASK	ANALYSIS + EVALUATION
RQ1 Who changes first?	Relevance + emotion → S by role; public E	CCF / conditional VAR → Lags + held-out predictive gain
RQ2 What changes at events?	Event links + emotion → S / E; common windows	Segmented regression / ITS → Level / slope + window checks
RQ3 How is fear explained?	Relations + topics → Cause / blame / duty	Relations + topic interpretation → Span P/R + source/period checks

Shared safeguards: fixed units + weights · source/time validation · original context

Figure 3. Research questions and computational evidence. Temporal comparisons use common bins; RQ3 retains passage-level evidence.

S measures attention; E measures emotion within relevant discourse. $B = S \times E$ is derived, not independent evidence. Prediction does not establish causation.

3.2 Evidence-driven implementation

The common-window pilot tests ingestion, role/date assignment, all three NLP modules and aggregation. Select by source overlap and independent dates. Expansion depends on measured processing and annotation costs.

Freeze outcomes, weights, labels, lag ranges and event windows before final comparisons.

Figure 5 shows the shared pipeline; secondary measures support interpretation.

3.3 Corpus construction and sampling

Table 2 lists candidates; access and sample extraction must verify obtainable coverage.

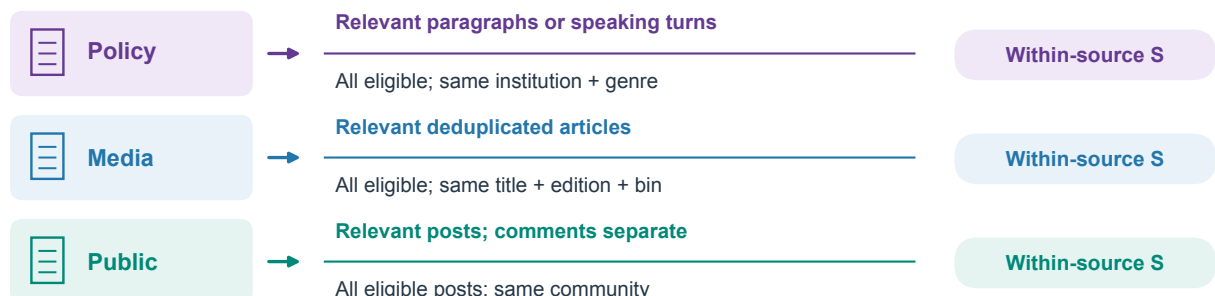
Table 2. Sources and coverage.

Candidate source	Role / type	Time coverage to establish	Access state
EPA, White House; UK/EU/AU/NZ records	Policy / institutional	Target 1988-2026; archive-specific gaps	● Candidate
UNFCCC; IPCC reports	Policy decisions; science	Publication-specific; discontinuous	● Candidate
NYT, Guardian, WSJ, FT; AU/NZ news	Media / articles	Title, edition and year entitlement	● Candidate
Letters, Usenet, preserved forums	Public / historical	Surviving dated records; no continuity assumed	● Candidate
Reddit research route	Public / platform	Rolling 5 years; 6-month delay (Reddit, n.d.)	● Candidate
Facebook, YouTube, Bluesky, Mastodon; forums	Public / platform	API/archive-specific; overlap unverified	● Candidate

Status: ● verified access/sample; ● candidate; ○ unresolved. Record advertised, accessible and acquired dates, gaps and release conditions.

Three source roles, three auditable denominators

General attention comes from the eligible collection, not climate-only retrieval.



Climate-only subsets support emotion and relations; they cannot estimate general attention.

Figure 4. Attention denominators. Numerator and denominator share an eligible collection; posts and comments remain separate.

Issue-independent sampling estimates attention; enriched climate retrieval supports emotion/relations. Prevalence requires known inclusion weights. Compare fixed stratum weights (stable source mix) with size weights (eligible population).

Keep historical public channels separate unless overlap supports comparison. IPCC retains a scientific subtype; specialist collections cannot supply government-wide denominators. Distinguish missing bins from zeros. Retain IDs, publication/collection dates, parent documents, roles, speakers, location evidence and duplicate clusters.

3.4 Operational definitions and construct validity

The primary outcome is future-oriented warming fear or worry in public texts. Fear, anxiety and worry are distinguishable, overlapping expressions, not diagnoses. Fear can concern anticipated threat; worry involves possible consequences. Collect immediate heat danger and long-term concern together, then label their horizons separately.

Table 3a. Constructs and measures.

Construct	Operational measure	Interpretation boundary
Attention S	Weighted relevant share of eligible units	Sampled output, not audience exposure
Emotion E	Validated label/share within relevant units	Expression, not clinical severity
Joint share B	Relevant emotional units / eligible units	Derived from S and E
Risk framing	Validated multi-label prototype/classifier scores	Mixed/other frames retained
Responsibility	Evidence-linked cause, blame and duty	Grammatical subject is not culpability
Stance	Position on a stated threat or policy	Policy opposition is not denial

In Equations (1)-(3), U is the eligible unit set for source stratum r and time bin t ; i indexes units, w is sampling weight, R indicates warming relevance and F with superscript future indicates fear/worry about anticipated warming. Units, weights and inclusion rules are identical.

$$S_{rt} = \frac{\sum_{i \in U_{rt}} w_i R_i}{\sum_{i \in U_{rt}} w_i} \quad (1)$$

$$E_{rt} = \frac{\sum_{i \in U_{rt}} w_i R_i F_i^{\text{future}}}{\sum_{i \in U_{rt}} w_i R_i} \quad (2)$$

$$B_{rt} = \frac{\sum_{i \in U_{rt}} w_i R_i F_i^{\text{future}}}{\sum_{i \in U_{rt}} w_i} = S_{rt} E_{rt} \quad (3)$$

S measures discussion share, E its emotional composition and B their joint prevalence. B will not enter a model as independent evidence alongside S and E . Empty denominators are missing. Scores require calibration or score-based interpretation; classifier probabilities do not establish psychological intensity.

Freeze weights and aggregation within comparisons. Article and policy-paragraph shares support within-series change and temporal comparison, but their levels cannot rank which role is more fearful.

3.4 Operational definitions (continued)

Retain each passage's target, horizon, source role, quoted speaker, emotion-holder and assertion/quotation scope. A newspaper reporting a scientist describing residents' fear contains three distinct positions. Risk statements need not express institutional emotion; unknown/multiple holders remain explicit. Quoted public emotion stays in its original source series.

Attach negation and modality to the expression they modify. "Should not panic" does not establish panic; a conditional future event may express present worry. Distinguish counterfactual and denied emotion, retain uncertainty for review, and avoid forcing ambiguous cases into positive labels.

Table 3b. Annotation boundaries (synthetic examples).

Synthetic passage	Emotion / target / horizon	Speaker, holder and scope
I worry about my children's future.	Worry; target unspecified; future	Self-expression; climate context needed
I fear warming will harm my children.	Fear; warming impacts; future	Self-expression; anticipated harm
The government said people should not panic.	Reassurance; no observed fear established	Government reported; negated advice
Scientists warn heatwaves are deadly.	Risk statement; heat; unspecified time	Reported warning; no fear required
If warming worsens, I fear our town will become unsafe.	Fear; warming impacts; future	Present fear; conditional consequence
I am not afraid of warming; I oppose this tax.	Fear denied; policy opposition	Negation of fear; distinct stance

Code horizon (immediate/near-term or longer/intergenerational) and threatened group (self/family/community) independently; retain unspecified categories. The guide records evidence spans and counterexamples, with separate development and held-out checks. These are text labels, not a clinical taxonomy.

Use multiple relevance prototypes and development-set thresholds. Frame directions need not be orthogonal; absolute projection can conflate opposite directions. Validate continuous measures as well as classifiers. Fear/anxiety ratios and stance bimodality remain secondary: denominator instability and classification mixtures can distort them; bimodality alone does not establish population polarisation.

For RQ3, link emotion changes to the feared outcome, affected group and expressed cause, blame or duty. Unexpressed attribution stays missing, regardless of nearby events.

3.5 Representation and three NLP components

Relevance detection feeds three NLP components (Figure 5). We compare lexical/TF-IDF retrieval with Sentence-BERT using precision, recall, F1 and missed cases (Reimers & Gurevych, 2019); freeze versioned, context-linked embeddings within comparisons. We report throughput (documents/hour), peak memory, runtime and annotation cost. Licence and local compatibility govern selection; MPS remains conditional. Shared representation does not imply joint training.

A modular NLP pipeline with shared representations

Relevance: lexical/TF-IDF versus Sentence-BERT; freeze the encoder after validation.

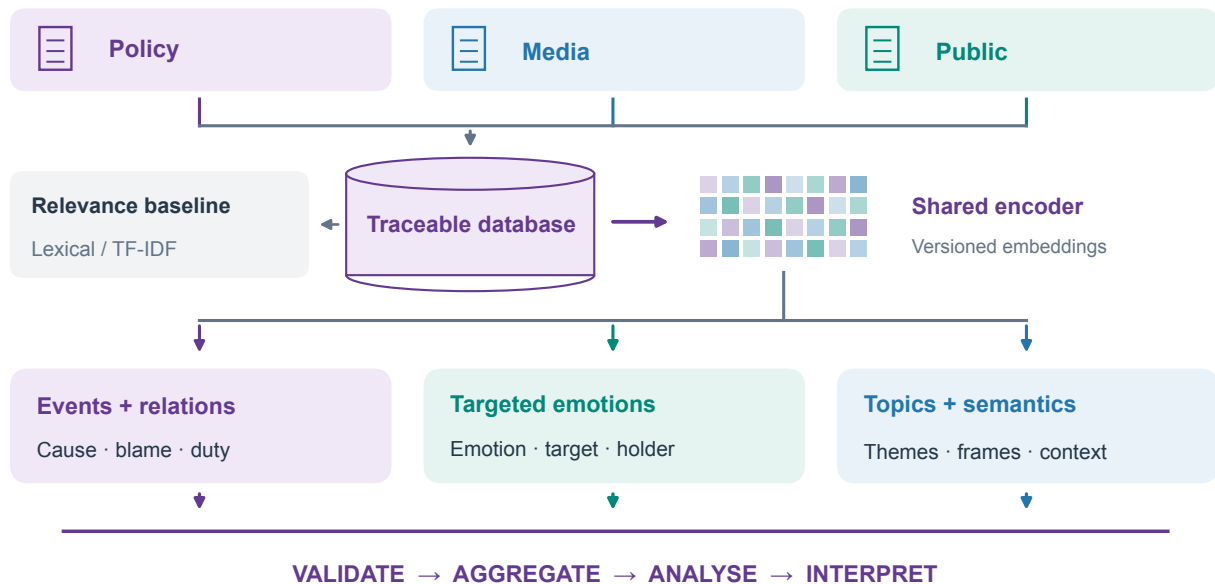


Figure 5. NLP pipeline. Context and provenance accompany all three components; validation precedes interpretation.

Event and relation extraction. Entities and parsing identify candidates for evaluated semantic role labelling (SRL) or contextual extraction. Outputs distinguish event, speaker, cause, blame, duty and affected group (Palmer et al., 2005; Xia & Ding, 2019), retaining evidence spans and scope. Verify analytical event dates independently.

Aspect-specific sentiment and emotion. We compare a GoEmotions-derived candidate with interpretable cues (Demszky et al., 2020). Identify the evaluated object (aspect), fear and project-defined future worry. Validate institutional risk versus holder-expressed emotion; nervousness is not automatically climate anxiety.

Temporal topics and semantics. We use BERTopic with shared embeddings (Grootendorst, 2022). Compare topic prevalence, frames and representative passages across roles/time; check resampling stability and outlier coverage. Interpret semantic change only with matched-source context and validated distinctions.

3.6 Temporal alignment and analysis

Use common coverage and frequency: monthly, daily/weekly for dense heat pilots, or quarterly for sparse institutions. Recompute ratios from counts; never duplicate quarters into months. GISTEMP describes regional climate; station/ERA5 data address exposure ([European Centre for Medium-Range Weather Forecasts, n.d.](#); [NASA Goddard Institute for Space Studies, n.d.](#)). Record location evidence, baselines and versions; publication location is not exposure.

RQ1: ordering and conditional prediction. Address trends, seasonality and serial dependence before CCF analysis. Equation (4) correlates processed series X and Y; positive k means X precedes Y. Compare role attention or policy/media attention with public emotion.

$$C_{XY}(k) = \text{Corr}(X_t, Y_{t+k}) \quad (4)$$

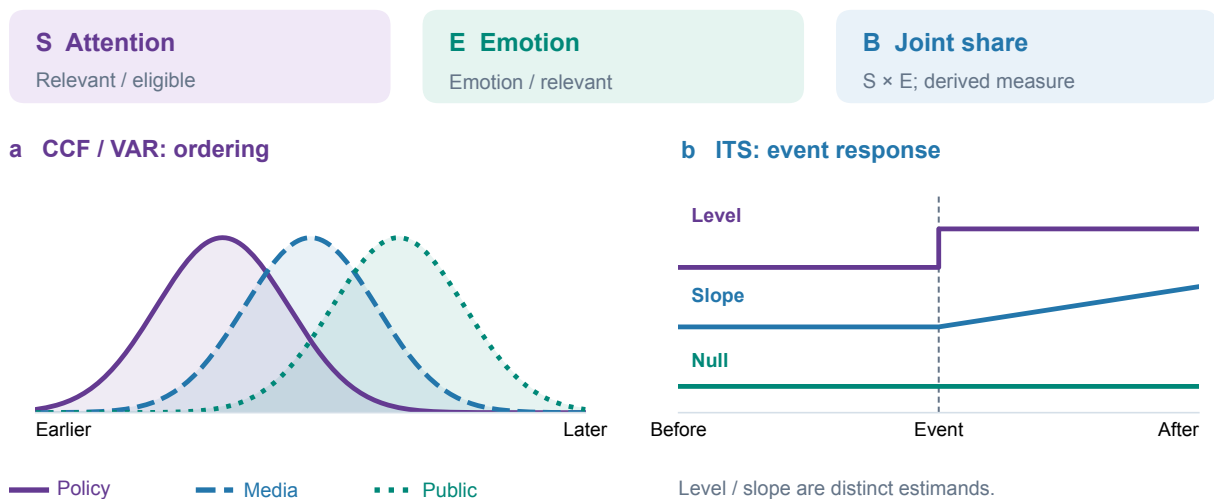
Where coverage and diagnostics permit, a small vector autoregression (VAR) tests added prediction beyond the outcome's own history ([Shojaie & Fox, 2022](#)). Report direction, lag range, uncertainty and held-out gain. CCF peaks alone do not select lag order. Compare role-specific S and public E; B remains secondary.

RQ2: event-associated change. Segmented regression/interrupted time series (ITS) estimates level and slope changes ([Lopez Bernal et al., 2017](#)). In Equation (5), t is centred on onset and D switches from 0 to 1; β_2 and β_3 estimate level and slope changes. Diagnose serial dependence in u and adjust uncertainty.

$$Y_t = \beta_0 + \beta_1 t + \beta_2 D_t + \beta_3 (tD_t) + u_t \quad (5)$$

Measure the discourse, then test its timing

Illustrative trajectories only; no observed data or estimated effects.



Inputs: aligned S / E series and event dates. Outputs: lags, predictive gain and level / slope.

Figure 6. Temporal analysis. All curves are schematic, not observed results.

Event selection. Select independently dated events before inspecting outcomes; require common coverage and auditable denominators. Pilot one event, distinguishing announcement, adoption, implementation and scientific release.

3.6 Event selection and narrative interpretation (continued)

Table 4. Candidate events.

Candidate / type	External date	Window / outcome	Comparison and limit
Kyoto adoption / policy	11 Dec 1997	Initially ±6 months; S/E level and slope	Historical overlap unverified
Paris adoption / policy	12 Dec 2015	Initially ±6 months; S/E level and slope	Anticipation; concurrent coverage
Heat episode / physical	○ Independent onset, peak and end	Dense local window; danger and future worry	Matched exposure; seasonality
IPCC release / scientific	○ Report-specific release date	Common coverage; S/E and frames	Shared shock; media spillover

Adoption dates follow UNFCCC records (United Nations Framework Convention on Climate Change, n.d.-a, n.d.-b). Extend initial six-month windows if trends, seasonality or serial dependence require longer baselines. Fix windows and aggregation before final tests using coverage and baseline adequacy.

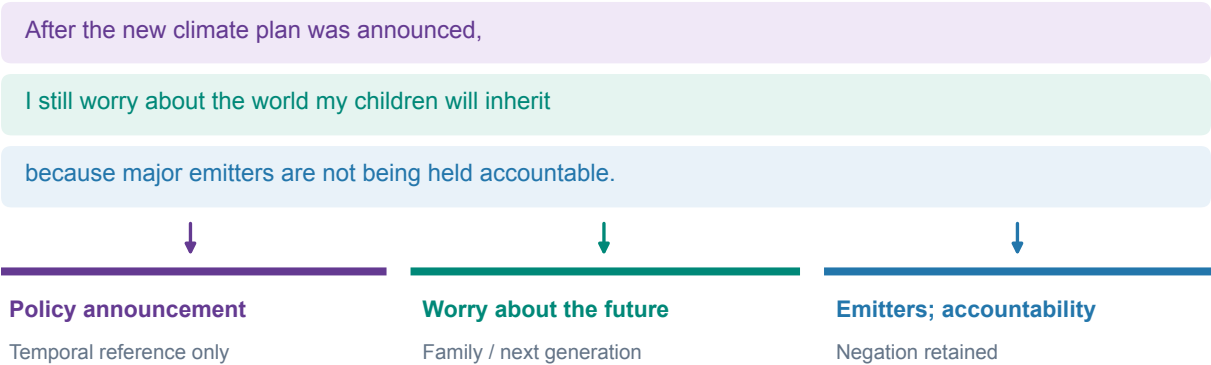
Controls require comparable pre-event measurement and defensibly different exposure. Register anticipation/concurrent shocks; inseparable events share a window. Placebo dates and alternative windows assess sensitivity, not causal identification (Hausman & Rapson, 2018).

Channel-transition diagnostic. Apply Equation (5) to independently documented corpus/channel transitions (e.g., letters coverage ending, forum uptake, Reddit eligibility), using comparable observations on both sides. Check S/E level and slope changes against fixed-source subsets. Coincident jumps flag possible composition bias, not proof of artificial emotion change. Without comparable overlap, document the coverage break and analyse strata separately.

RQ3: narrative explanation. Relations and topics locate causes, blame, duties and threatened futures. Check quotation and stance before interpreting adoption, contestation or reinterpretation. Narrative attribution differs from estimated event effects; retain representative and contradictory passages.

One passage, several linked pieces of evidence

Synthetic example; contextual relations are not verified causal effects.



Still unresolved: policy caused the worry; who has the duty to act.
Figure 7. Evidence-linked annotation. Synthetic text illustrates labels and attribution; timing does not establish causation.

3.7 Evaluation and validity

A role-balanced development batch of approximately 150 passages identifies ambiguities and annotation costs, not final performance. Size the separate stratified test set using prevalence, cost and desired precision; disclose rare-label enrichment.

We will seek an independent second reader for a subset. Without one, report separated repeat assessment and disagreements as limited single-author validation, not inter-rater reliability. Split development/test sets by document and duplicate cluster; use source/period holdouts to assess transfer.

Table 5. Computational evaluation plan.

Task	Baseline / reference	Candidate / comparison	Metric / output	Validation setting
Relevance	Lexical / TF-IDF	Sentence-BERT retrieval	Precision, recall, F1	Source/period holdout; background misses
Emotion	Interpretable cues	GoEmotions-derived candidate	Macro/per-label F1; calibration	Annotated targets/holders; source/period transfer
Relations	Predicate/entity rules	SRL + contextual relations	Span and relation P/R	Annotated evidence; source/period transfer
Topics	Matched-source reference	BERTopic configurations	Stability, coverage; coherence review	Resampled texts; contextual passages
RQ1	Own-history model	Conditional VAR + roles	Held-out MSE gain; lag uncertainty	Common window; chronological holdout
RQ2	Pre-event trend	Segmented regression / ITS	Level/slope; uncertainty	Event windows; placebo dates
RQ3	Annotation guide	Relations + topic evidence	Agreement; traceable spans	Quotation/stance checks; counterexamples

Use identical eligible test texts and documented configurations. P/R = precision/recall; MSE = mean squared error. Audit dates, roles, duplicates and coverage; contextual evidence supports topic/RQ3 interpretation.

Development $F1 < 0.70$ or two-rater $\kappa < 0.65$ flags review; these provisional thresholds do not validate aggregate measures. Resampling/error perturbations will test classification-error effects on temporal estimates. Account for time-block, source, author and duplicate dependence in uncertainty.

Freeze outcomes, thresholds, weights and lag ranges before final comparisons; label alternatives as sensitivity/exploratory analyses. Non-significance proves neither absent physical influence nor exclusive policy control (Wasserstein & Lazar, 2016).

3.8 Project risks and scope adjustment

Database construction is the main delivery risk: delays consume validation and analysis time. Table 6 links observable triggers to specific fallbacks. Ratings indicate management priority, not estimated probabilities; Section 3.10 covers ethics and OHS.

3.8.1 Risk assessment

Table 6. Risks, triggers and fallbacks.

Risk	Trigger / consequence	Rating	Concrete fallback
R1. Database overrun	Core freeze missed: analysis time shrinks.	High	At 18 December, stop optional ingestion; freeze validated batches and reduce source/window breadth.
R2. Access or release	Access denied, delayed or incompatible with reuse.	High	If Reddit is unavailable at the pilot gate, assess a permitted Mastodon community; restrict comparison to its verified common window.
R3. Sparse coverage	Low effective counts or insufficient common bins.	High	Use common non-overlapping quarters; otherwise shorten the window. Sparse historical texts remain contextual evidence.
R4. NLP validity	Source/period audit misses prespecified quality targets.	High	Refine labels and revalidate; narrow unreliable distinctions. Report unsupported constructs qualitatively rather than substitute unvalidated scores.
R5. Temporal validity	Diagnostics fail or events overlap; estimates unstable.	High	Reduce lag/model complexity or retain descriptive CCF/event profiles with uncertainty. Null results do not establish another driver.
R6. Composition drift	Source mix or encoder changes produce a break.	High	Use the channel-transition diagnostic (3.6), fixed-source subsets and a frozen encoder; separate incompatible periods.
R7. Compute / review	Pilot throughput or reviewer availability misses budget.	Medium	Cache batches, cap comparisons and use CPU inference; reduce validation sample breadth and disclose any single-reader limitation.
R8. Disruption / loss	Illness, corruption or failed restore threatens delivery.	Medium	Restore tested, versioned backups; prioritise core analyses with the supervisor and use protected contingency.

3.8.2 Review points and scope adjustment

Mastodon is a candidate fallback, not a guaranteed archive: instance permissions, dated coverage and an all-topic denominator must be verified ([Mastodon, n.d.](#)). Permitted letters or forum archives may support earlier public strata. These sources will not be spliced into a continuous population series. If no public source qualifies, narrow the three-role comparison to a supported window; document any RQ left unanswered.

Review gates are 16 October (pilot), 18 December (core corpus), 15 January (validation) and 28 February 2027 (all analyses). A missed gate or sustained workload above 10-20 hours/week triggers scope review. Protect the three roles, three NLP components and validation. Counts below 100 flag review; effective counts, uncertainty and usable time points determine aggregation.

3.9 Milestones, resources and time allocation

At 10-20 hours/week, the provisional budget is 450-590 hours plus 15% contingency. All analysis and robustness checks must finish by 28 February 2027. The 320-400 hours allocated before March require about 14-17 hours/week before contingency; sustained lower capacity will trigger early scope reduction.

A research calendar with protected analysis time

10–20 hours/week · planned task effort 450–590 h + 15% contingency

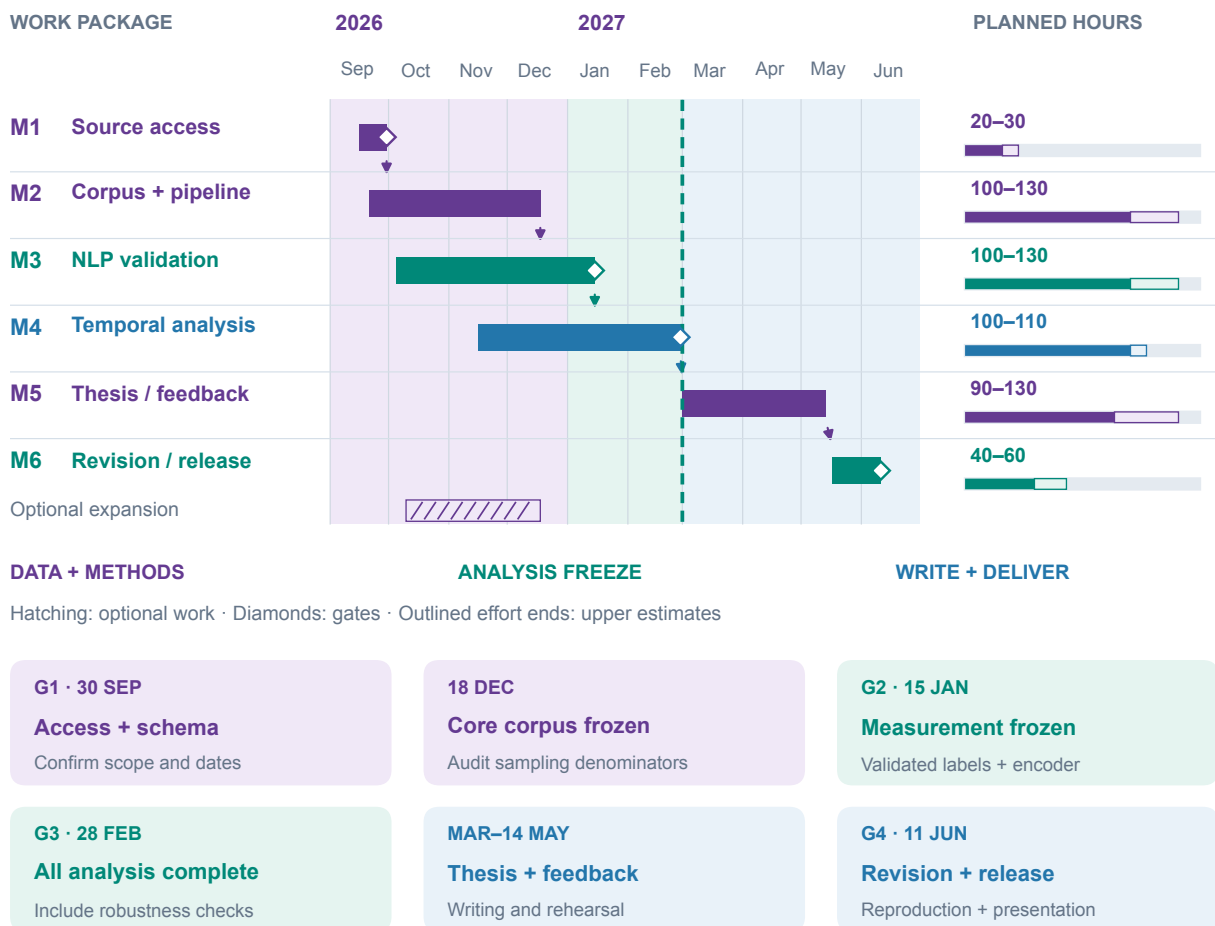


Figure 8. Work programme and effort. Solid/outlined bars show lower/upper estimates; hatching denotes optional work. Hours are planned.

Resource dependencies are permitted archives, storage/local compute, annotation time and supervisory feedback. The pilot is due by 16 October; core data and denominator checks freeze on 18 December. Measurement validation freezes on 15 January, followed by all temporal, event, topic and attribution analyses by 28 February. Release requires a permissions review and reproduction instructions.

Assessment anchors: seminar, 12-16 October; Thesis Plan, 1-19 March 2027; rehearsal, 10-14 May; final pitch/poster/demonstration/Q&A, 7-11 June ([The University of Queensland \[UQ\], 2026](#)). March-June centres on writing and feedback using completed analyses.

Missed freezes trigger narrower comparisons while preserving three roles, three modules and the February target.

3.10 OHS, ethics and data governance

This computer-based documentary study involves no planned fieldwork or participant recruitment. Workstation, data-handling and ethics requirements will be reviewed with the supervisor before relevant work.

Table 7. OHS and data controls.

Issue	Preventive control	Review / residual issue
Posture and repetitive work	Ergonomic setup, breaks and task rotation	Adjust if discomfort develops.
Workload / distressing text	Bounded sessions and annotation batches; university support	Monitor fatigue and distress.
Personal data / quotations	Minimise identifiers; restrict raw access; assess quotation reidentification	Public access is not blanket consent.
Group representation	Use observable source roles; retain uncertainty	Do not infer sensitive identities or population prevalence.
Services / release	Review licences, transfer, retention and deletion; favour local processing	Features and embeddings may remain restricted.

Supervisory review. Before accessing human-authored datasets, Dai Pan will obtain Mashhuda Glencross's documented review of permissions, identifiability, storage and release. This project gate is separate from formal clearance. UQ students cannot be Chief Investigator (CI); the supervisor will submit as CI where appropriate (UQ, n.d.-c).

UQ determination. Seek advice from Research Ethics and Integrity (humanethics@research.uq.edu.au) on research involving people or their data, and document the outcome. Public availability does not establish exemption; intended publication cannot rely on the teaching-only exception. Apply through MyResearch for lower-risk or Human Research Ethics Committee (HREC) review, or register an eligible exemption. Relevant work begins only after clearance or exemption conditions are established (UQ, n.d.-a, n.d.-b). No approval is claimed here.

Later human evaluation. If human A/B evaluation is proposed, the supervisor will first review the protocol, recruitment, consent, tasks and data plan. Whether reviewers become research participants will be checked with Research Ethics and Integrity. Any required new approval or amendment must precede recruitment and data collection; a supervisor's quality check is not institutional approval (UQ, n.d.-a).

Restricted originals remain separate from permitted derivatives. Storage, external inference and release require source-specific checks of permissions, retention and deletion conditions.

3.11 Outputs, interpretation and research record

The database will record source-time keys, permitted IDs/URLs, weights, missingness, climate metadata, model versions, features and uncertainty. Permitted CSV/Parquet exports and code will include a data dictionary and versioned configuration; restricted data will have access descriptions.

The thesis will connect temporal findings to validated passages, including contradictory evidence and limits of population inference. A decision log, source register and progress notes will support supervisory discussion. Reproducible evidence will underpin subsequent publication or algorithmic claims.

References

- Albrecht, G., Sartore, G.-M., Connor, L., Higginbotham, N., Freeman, S., Kelly, B., Stain, H., Tonna, A., & Pollard, G. (2007). Solastalgia: The distress caused by environmental change. *Australasian Psychiatry*, 15(Suppl. 1), S95–S98. <https://doi.org/10.1080/10398560701701288>
- Brulle, R. J., Carmichael, J., & Jenkins, J. C. (2012). Shifting public opinion on climate change: An empirical assessment of factors influencing concern over climate change in the U.S., 2002–2010. *Climatic Change*, 114(2), 169–188. <https://doi.org/10.1007/s10584-012-0403-y>
- Callendar, G. S. (1938). The artificial production of carbon dioxide and its influence on temperature. *Quarterly Journal of the Royal Meteorological Society*, 64(275), 223–240. <https://doi.org/10.1002/qj.49706427503>
- Clayton, S. (2020). Climate anxiety: Psychological responses to climate change. *Journal of Anxiety Disorders*, 74, Article 102263. <https://doi.org/10.1016/j.janxdis.2020.102263>
- Demszky, D., Movshovitz-Attias, D., Ko, J., Cowen, A., Nemade, G., & Ravi, S. (2020). GoEmotions: A dataset of fine-grained emotions. In D. Jurafsky, J. Chai, N. Schluter, & J. Tetreault (Eds.), *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics* (pp. 4040–4054). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2020.acl-main.372>
- Entman, R. M. (1993). Framing: Toward clarification of a fractured paradigm. *Journal of Communication*, 43(4), 51–58. <https://doi.org/10.1111/j.1460-2466.1993.tb01304.x>
- European Centre for Medium-Range Weather Forecasts (n.d.). *ERA5 post-processed daily statistics on single levels from 1940 to present*. Retrieved September 15, 2026, from <https://www.ecmwf.int/en/forecasts/datasets/era5-post-processed-daily-statistics-single-levels-1940-present>
- Google (n.d.). *Gemini* [Large language model]. Retrieved September 15, 2026, from <https://gemini.google.com/>
- Grootendorst, M. (2022). *BERTopic: Neural topic modeling with a class-based TF-IDF procedure*. arXiv. <https://doi.org/10.48550/arXiv.2203.05794>
- Hamaker, E. L., Kuiper, R. M., & Grasman, R. P. P. P. (2015). A critique of the cross-lagged panel model. *Psychological Methods*, 20(1), 102–116. <https://doi.org/10.1037/a0038889>
- Hausman, C., & Rapson, D. S. (2018). Regression discontinuity in time: Considerations for empirical applications. *Annual Review of Resource Economics*, 10(1), 533–552. <https://doi.org/10.1146/annurev-resource-121517-033306>
- Intergovernmental Panel on Climate Change (n.d.). *History of the IPCC*. Retrieved September 15, 2026, from <https://www.ipcc.ch/about/history/>
- Lopez Bernal, J., Cummins, S., & Gasparrini, A. (2017). Interrupted time series regression for the evaluation of public health interventions: A tutorial. *International Journal of Epidemiology*, 46(1), 348–355. <https://doi.org/10.1093/ije/dyw098>
- Mastodon (n.d.). *Timelines API methods*. Retrieved September 15, 2026, from <https://docs.joinmastodon.org/methods/timelines/>
- McCombs, M. E., & Shaw, D. L. (1972). The agenda-setting function of mass media. *Public Opinion Quarterly*, 36(2), 176–187. <https://doi.org/10.1086/267990>
- NASA Goddard Institute for Space Studies (n.d.). *GISS surface temperature analysis (GISTEMP v4)*. Retrieved September 15, 2026, from <https://data.giss.nasa.gov/gistemp/>
- O'Neill, S., & Nicholson-Cole, S. (2009). “Fear won’t do it”: Promoting positive engagement with climate change through visual and iconic representations. *Science Communication*, 30(3), 355–379. <https://doi.org/10.1177/1075547008329201>
- OpenAI (n.d.-a). *ChatGPT* [Large language model]. Retrieved September 15, 2026, from <https://chatgpt.com/>

References (continued)

- OpenAI (n.d.-b). *Codex* [Computer software]. Retrieved September 15, 2026, from <https://openai.com/codex/>
- Palmer, M., Gildea, D., & Kingsbury, P. (2005). The Proposition Bank: An annotated corpus of semantic roles. *Computational Linguistics*, 31(1), 71–106. <https://doi.org/10.1162/0891201053630264>
- Pihkala, P. (2022). Toward a taxonomy of climate emotions. *Frontiers in Climate*, 3, Article 738154. <https://doi.org/10.3389/fclim.2021.738154>
- Pontiki, M., Galanis, D., Pavlopoulos, J., Papageorgiou, H., Androutsopoulos, I., & Manandhar, S. (2014). SemEval-2014 task 4: Aspect based sentiment analysis. In P. Nakov & T. Zesch (Eds.), *Proceedings of the 8th International Workshop on Semantic Evaluation (SemEval 2014)* (pp. 27–35). Association for Computational Linguistics. <https://doi.org/10.3115/v1/S14-2004>
- Reddit (n.d.). *Reddit for Researchers program*. Retrieved September 15, 2026, from <https://support.reddithelp.com/hc/en-us/articles/49381918834964-Reddit-for-Researchers-Program>
- Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In K. Inui, J. Jiang, V. Ng, & X. Wan (Eds.), *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)* (pp. 3982–3992). Association for Computational Linguistics. <https://doi.org/10.18653/v1/D19-1410>
- Shojaie, A., & Fox, E. B. (2022). Granger causality: A review and recent advances. *Annual Review of Statistics and Its Application*, 9(1), 289–319. <https://doi.org/10.1146/annurev-statistics-040120-010930>
- United Nations Framework Convention on Climate Change (n.d.-a). *The Kyoto Protocol*. Retrieved September 15, 2026, from <https://www.unfccc.int/process-and-meetings/the-kyoto-protocol>
- United Nations Framework Convention on Climate Change (n.d.-b). *The Paris Agreement*. Retrieved September 15, 2026, from <https://unfccc.int/process-and-meetings/the-paris-agreement>
- The University of Queensland (n.d.-a). *Ethics application*. Retrieved September 15, 2026, from <https://research-support.uq.edu.au/resources-and-support/ethics-integrity-and-compliance/human-ethics/ethics-application>
- The University of Queensland (n.d.-b). *Human research ethics procedure*. Retrieved September 15, 2026, from <https://policies.uq.edu.au/document/view-current.php?id=346>
- The University of Queensland (n.d.-c). *MyResearch frequently asked questions*. Retrieved September 15, 2026, from <https://research-support.uq.edu.au/resources-and-support/research-systems/myresearch/myresearch-frequently-asked-questions>
- The University of Queensland (2026). *Research and development methods and practice (REIT7842): Semester 2, 2026*. <https://course-profiles.uq.edu.au/course-profiles/REIT7842-60234-7660#assessment>
- Wasserstein, R. L., & Lazar, N. A. (2016). The ASA statement on p-values: Context, process, and purpose. *The American Statistician*, 70(2), 129–133. <https://doi.org/10.1080/00031305.2016.1154108>
- Xia, R., & Ding, Z. (2019). Emotion-cause pair extraction: A new task to emotion analysis in texts. In A. Korhonen, D. Traum, & L. Màrquez (Eds.), *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics* (pp. 1003–1012). Association for Computational Linguistics. <https://doi.org/10.18653/v1/P19-1096>
- Zhou, S., Yang, X., Wang, Y., Zheng, X., & Zhang, Z. (2023). Affective agenda dynamics on social media: Interactions of emotional content posted by the public, government, and media during the COVID-19 pandemic. *Humanities and Social Sciences Communications*, 10(1), Article 797. <https://doi.org/10.1057/s41599-023-02265-x>